```
> restart;# GravBend compile time 27 sec
```

```
> de := diff(g(r), r$2) = r·g(r)/R;
```

$$de := \frac{d^2}{dr^2}\, g(r) = \frac{r\, g(r)}{R} \tag{1}$$

```
> ics := g(2·R) = 0, D(g)(0) = R;
```

$$ics := g(2\,R) = 0,\ D(g)(0) = R \tag{2}$$

```
> sol := dsolve({de, ics});
```

$$sol := g(r) = \frac{2\,R\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{R}\right)^{1/3} R\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R}\right)^{1/3} r\right)}{\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{R}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R}\right)^{1/3} R\right) + \mathrm{AiryBi}\left(-2\left(-\frac{1}{R}\right)^{1/3} R\right) 3^{1/6}\right)} - \frac{2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R}\right)^{1/3} R\right) R\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{R}\right)^{1/3} r\right)}{\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{R}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R}\right)^{1/3} R\right) + \mathrm{AiryBi}\left(-2\left(-\frac{1}{R}\right)^{1/3} R\right) 3^{1/6}\right)} \tag{3}$$

```
> de1 := diff(g1(r), r$2) = r·g1(r)/(R·b);
```

$$de1 := \frac{d^2}{dr^2}\, g1(r) = \frac{r\, g1(r)}{R\, b} \tag{4}$$

```
> ics1 := g1(2·R·b) = 0, D(g1)(0) = R·b;
```

$$ics1 := g1(2\,R\,b) = 0,\ D(g1)(0) = R\,b \tag{5}$$

```
> sol := dsolve({de1, ics1});
```

$$sol := g1(r) \tag{6}$$

$$= \frac{2\,R\,b\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3} R\,b\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3} r\right)}{\left(-\frac{1}{R\,b}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3} R\,b\right) + \mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3} R\,b\right) 3^{1/6}\right)} - \frac{2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3} R\,b\right) R\,b\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3} r\right)}{\left(-\frac{1}{R\,b}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3} R\,b\right) + \mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3} R\,b\right) 3^{1/6}\right)}$$

```
> g1(r) :=
  2·R·b·π·AiryBi(-2·(-1/(R·b))^(1/3)·R·b)·AiryAi(-(-1/(R·b))^(1/3)·r)
  /((-1/(R·b))^(1/3)·Γ(2/3)·(3^(2/3)·AiryAi(-2·(-1/(R·b))^(1/3)·R·b)
  + AiryBi(-2·(-1/(R·b))^(1/3)·R·b)·3^(1/6)))
  - (2·AiryAi(-2·(-1/(R·b))^(1/3)·R·b)·R·b·π·AiryBi(-(-1/(R·b))^(1/3)·r))
  /((-1/(R·b))^(1/3)·Γ(2/3)·(3^(2/3)·AiryAi(-2·(-1/(R·b))^(1/3)·R·b)
  + AiryBi(-2·(-1/(R·b))^(1/3)·R·b)·3^(1/6)));
```

$$g1 := r \tag{7}$$

$$\mapsto \left(2\cdot R\cdot b\cdot \pi\cdot \mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)\cdot \mathrm{AiryAi}\left(-\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot r\right)\right) \Big/ \left(\left(-\frac{1}{R\cdot b}\right)^{1/3}\right.$$

$$\cdot\Gamma\left(\frac{2}{3}\right)\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)\cdot 3^{1/6}\right)\Bigg)$$
$$-\left(2\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)\cdot R\cdot b\cdot\pi\cdot\mathrm{AiryBi}\left(-\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot r\right)\right)\Big/\left(\left(-\frac{1}{R\cdot b}\right)^{1/3}\right.$$
$$\cdot\Gamma\left(\frac{2}{3}\right)\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)\cdot 3^{1/6}\right)\Bigg)$$

> `simplify(sol);`

$$-\left(2\,R\,b\,\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}r\right)-\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}r\right)\right)\right)\Big/\left(\left(-\frac{1}{R\,b}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)\right)=-\left(2\,R\,b\,\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}r\right)-\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}r\right)\right)\right)\Big/\left(\left(-\frac{1}{R\,b}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)\right) \tag{8}$$

> `change_of_vars := { r=b·rho, g(r)=f(rho) };`

$$change_of_vars := \{r=b\,\rho,\ g(r)=f(\rho)\} \tag{9}$$

> `tmp := PDEtools:-dchange(change_of_vars, [de1, ics1], [g(rho), rho]);`

$$tmp := \left[-\frac{2\,R\,b^2\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\left(-\frac{1}{R\,b}\right)^{2/3}\rho\,\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)}{\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)}\right.$$
$$+\frac{2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)R\,b^2\,\pi\left(-\frac{1}{R\,b}\right)^{2/3}\rho\,\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)}{\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)}$$
$$=\frac{1}{R}\left(\rho\left(\left(2\,R\,b\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)\right)\Big/\left(\left(\right.\right.\right.\right.$$
$$\left.\left.\left.\left.-\frac{1}{R\,b}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)\right)\right.$$
$$\left.-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)R\,b\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)\right)\Big/\left(\left(-\frac{1}{R\,b}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{Ai}\right.\right.\right.$$

$$^{3}\,R\,b\Big)\,3^{1/6}\Big)\Big)\Big)\Big),\,0=0,\,\frac{R\,b\,\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}}{3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}}$$

$$+\frac{\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,R\,b\,3^{2/3}}{3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}}=R\,b\Bigg]$$

> `simplify(tmp[1]);`

$$\Big(2\,\pi\Big(-\frac{1}{R\,b}\Big)^{2/3}R\,b^{2}\,\rho\Big(\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryBi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)-\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryAi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)\Big)\Big)\Big/\Big(\Big(3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}\Big)\,\Gamma\Big(\frac{2}{3}\Big)\Big)=-\Big(2\,\rho\,\pi\,b\Big(\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryBi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)-\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryAi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)\Big)\Big)\Big/\Big(\Big(-\frac{1}{R\,b}\Big)^{1/3}\Big(3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}\Big)\,\Gamma\Big(\frac{2}{3}\Big)\Big) \tag{11}$$

> `tmp[3];`

$$\frac{R\,b\,\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}}{3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}}+\frac{\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,R\,b\,3^{2/3}}{3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}}=R\,b \tag{12}$$

> `eq3 := (2*(-1/(R*b))^(2/3)*Pi*R*b^2*rho*(AiryAi(-2*(-1/(R*b))^(1/3)*R*b)*AiryBi(-(-1/(R*b))^(1/3)*b*rho)-AiryBi(-2*(-1/(R*b))^(1/3)*R*b)*AiryAi(-(-1/(R*b))^(1/3)*b*rho)))/((3^(2/3)*AiryAi(-2*(-1/(R*b))^(1/3)*R*b)+AiryBi(-2*(-1/(R*b))^(1/3)*R*b)*3^(1/6))*Gamma(2/3))=-(2*rho*Pi*b*(AiryAi(-2*(-1/(R*b))^(1/3)*R*b)*AiryBi(-(-1/(R*b))^(1/3)*b*rho)-AiryBi(-2*(-1/(R*b))^(1/3)*R*b)*AiryAi(-(-1/(R*b))^(1/3)*b*rho)))/((-1/(R*b))^(1/3)*(3^(2/3)*AiryAi(-2*(-1/(R*b))^(1/3)*R*b)+AiryBi(-2*(-1/(R*b))^(1/3)*R*b)*3^(1/6))*Gamma(2/3));`

$$eq3 := \Big(2\,\pi\Big(-\frac{1}{R\,b}\Big)^{2/3}R\,b^{2}\,\rho\Big(\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryBi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)-\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryAi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)\Big)\Big)\Big/\Big(\Big(3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)$$

$$+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}\Big)\,\Gamma\Big(\frac{2}{3}\Big)\Big)=-\Big(2\,\rho\,\pi\,b\Big(\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryBi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)-\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryAi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)\Big)\Big)$$

$$\Big/\Big(\Big(-\frac{1}{R\,b}\Big)^{1/3}\Big(3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}\Big)\,\Gamma\Big(\frac{2}{3}\Big)\Big);$$

$$eq3 := \Big(2\,\pi\Big(-\frac{1}{R\,b}\Big)^{2/3}R\,b^{2}\,\rho\Big(\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryBi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)-\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryAi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)\Big)\Big)\Big/\Big(\Big(3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)$$

$$+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}\Big)\,\Gamma\Big(\frac{2}{3}\Big)\Big)=-\Big(2\,\rho\,\pi\,b\Big(\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryBi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)-\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,\mathrm{AiryAi}\Big(-\Big(-\frac{1}{R\,b}\Big)^{1/3}b\,\rho\Big)\Big)\Big)$$

$$\Big/\Big(\Big(-\frac{1}{R\,b}\Big)^{1/3}\Big(3^{2/3}\,\mathrm{AiryAi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)+\mathrm{AiryBi}\Big(-2\Big(-\frac{1}{R\,b}\Big)^{1/3}R\,b\Big)\,3^{1/6}\Big)\,\Gamma\Big(\frac{2}{3}\Big)\Big) \tag{13}$$

$$+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)\Gamma\left(\frac{2}{3}\right)\Big)=-\Big(2\,\rho\,\pi\,b\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)-\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)\right)\Big)$$
$$\Big/\left(\left(-\frac{1}{R\,b}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)\Gamma\left(\frac{2}{3}\right)\right)$$

> $sol3 := solve(\{eq3, ics1\});$

$$sol3 := \{R = R,\ b = b,\ \rho = \rho\} \tag{14}$$

> $restart;$

> $b := 1980;\ R;$

$$b := 1980$$
$$R \tag{15}$$

> $g1(r) :=$

$$\frac{2\,R\,b\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}r\right)}{\left(-\frac{1}{R\,b}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)}$$
$$-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)R\,b\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}r\right)\right)\Big/\left(\left(-\frac{1}{R\,b}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(3^{2/3}\,\mathrm{Ai}\right.\right.$$
$$\left.\left.{}^{3}\,R\,b\right)3^{1/6}\right)\Big);$$

$$g1 := r \tag{16}$$
$$\mapsto \left(2\cdot R\cdot b\cdot\pi\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)\cdot\mathrm{AiryAi}\left(-\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot r\right)\right)\Big/\left(\left(-\frac{1}{R\cdot b}\right)^{1/3}\right.$$
$$\left.\cdot\Gamma\left(\frac{2}{3}\right)\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)\cdot 3^{1/6}\right)\right)$$
$$-\left(2\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)\cdot R\cdot b\cdot\pi\cdot\mathrm{AiryBi}\left(-\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot r\right)\right)\Big/\left(\left(-\frac{1}{R\cdot b}\right)^{1/3}\right.$$
$$\left.\cdot\Gamma\left(\frac{2}{3}\right)\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{R\cdot b}\right)^{1/3}\cdot R\cdot b\right)\cdot 3^{1/6}\right)\right)$$

> $eq3 := \left(2\left(-\frac{1}{R\,b}\right)^{2/3}\pi\,R\,b^2\,\rho\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)-\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)\right)\right)\Big/\left(\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\right.\right.$$
$$\left.\left.+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)3^{1/6}\right)\Gamma\left(\frac{2}{3}\right)\right)=-\Big(2\,\rho\,\pi\,b\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}\right.\right.$$
$$\left.\left.{}^{3}\,R\,b\right)\mathrm{AiryBi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)-\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{R\,b}\right)^{1/3}b\,\rho\right)\right)\Big)$$
$$\Big/\left(\left(-\frac{1}{R\,b}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}R\,b\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{R\,b}\right)^{1/3}\right.\right.$$
$$\left.\left.{}^{3}\,R\,b\right)3^{1/6}\right)\Gamma\left(\frac{2}{3}\right)\right);$$

$$eq3 := \left(7840800\left(-\frac{1}{1980\,R}\right)^{2/3}\pi\,R\,\rho\left(\mathrm{AiryAi}\left(-3960\left(-\frac{1}{1980\,R}\right)^{1/3}R\right)\mathrm{AiryBi}\left(-1980\left(-\frac{1}{1980\,R}\right)^{1/}\right.\right. \tag{17}$$

$$^{3}\rho\Big) - \mathrm{AiryBi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,\mathrm{AiryAi}\Big(-1980\left(-\frac{1}{1980\,R}\right)^{1/3}\rho\Big)\Big)\Big)\Big/\Big(\Big(3^{2/3}\,\mathrm{AiryAi}\Big($$

$$-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big) + \mathrm{AiryBi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,3^{1/6}\Big)\,\Gamma\Big(\frac{2}{3}\Big)\Big) =$$

$$-\Big(3960\,\rho\,\pi\,\Big(\mathrm{AiryAi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,\mathrm{AiryBi}\Big(-1980\left(-\frac{1}{1980\,R}\right)^{1/3}\rho\Big) - \mathrm{AiryBi}\Big($$

$$-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,\mathrm{AiryAi}\Big(-1980\left(-\frac{1}{1980\,R}\right)^{1/3}\rho\Big)\Big)\Big)\Big/\Big(\Big(-\frac{1}{1980\,R}\right)^{1/}$$

$$^{3}\Big(3^{2/3}\,\mathrm{AiryAi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big) + \mathrm{AiryBi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,3^{1/6}\Big)\,\Gamma\Big(\frac{2}{3}\Big)\Big)$$

> $ics2 := g1(2\cdot R\cdot b) = 0,\ \mathrm{D}(g1)(0) = R\cdot b;$

$$ics2 := 0 = 0,\ \frac{1980\,R\,\mathrm{AiryBi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,3^{1/6}}{3^{2/3}\,\mathrm{AiryAi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big) + \mathrm{AiryBi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,3^{1/6}} \tag{18}$$

$$+\ \frac{1980\,\mathrm{AiryAi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,R\,3^{2/3}}{3^{2/3}\,\mathrm{AiryAi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big) + \mathrm{AiryBi}\Big(-3960\left(-\frac{1}{1980\,R}\right)^{1/3} R\Big)\,3^{1/6}} = 1980\,R$$

> $sol3 := solve(\{eq3, ics2\});$

$$sol3 := \{R = R,\ \rho = \rho\} \tag{19}$$

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